CASH TO COMPANY

Item Bank

"Measure, improve, and operate smarter"

Per-unit quiz / test items · 4 units · ETS-quality alignment to Jump$tart, CEE, CCSS, C3, CASEL, NGSS-

Grade 8 — Cash to Company Item Bank

Year theme: *Measure, improve, and operate smarter.* This bank covers all four Grade 8 units (From idea to test · The break-even truth · Marketing as experiment · Operate smarter). Items use linear equations, A/B test analysis, and PDSA framing (CCSS 8.EE.C.7, 8.F.B.4, 8.SP.A.1). Distractors target the Grade 8 Common Misconceptions table. No family-money probes; "for a hypothetical" framing throughout. Cash's Grade 8 voice: "*A little less wrong, every week.*"

Unit 1: From idea to test

MC1. *"SCAMPER and How Might We are tools for..."*

A. Picking a winner · B. Generating distinct alternatives from a single problem · C. Pricing a product · D. Building a survey

Answer: B · **Standard:** *NGSS-ETS MS-ETS1-1 — Define the criteria and constraints of a design problem.* · **Strand outcome (PSIG, 8):** *I can run a structured ideation flow (problem → 10 ideas → 3 finalists → 1 selected) using at least 2 named techniques (SCAMPER, "How Might We").* · **Misconception targeted:** "*Ideation = guessing harder*" — corrects.

MC2. *"A 'value lever' a business pulls is..."*

A. A discount code · B. One of: better product, lower price, OR better experience · C. A logo redesign · D. A bigger ad budget

Answer: B · **Standard:** *CEE Standard 5 (Financial Investing), Grade 8 — Investors who own a corporation's stock or a corporate bond can earn a return.* · **Strand outcome (BAVC, 8):** *I can analyze a business case (1-2 pages) and identify which of three "value levers" (better product, lower price, better experience) it pulled and how.* · **Misconception targeted:** "*All marketing moves are 'levers'*" — corrects.

MC3. *"Of 10 ideas, you narrowed to 3 finalists. The right next step is..."*

A. Pick the cheapest · B. Pick the one you like most · C. Test each finalist against named criteria (customer fit, cost, time, fit-with-skills) · D. Toss a coin

Answer: C · **Standard:** *NGSS-ETS MS-ETS1-1.* · **Strand outcome (PSIG, 8):** *I can run a structured ideation flow.* · **Misconception targeted:** "*Picking is intuition*" — corrects.

SA1. *"Run a structured ideation: take the problem 'middle-schoolers run out of phone battery during 7th period.' Generate 10 ideas (you can list briefly), pick 3 finalists, select 1 with reasons."*

Model answer: "*10 ideas: 1) shared portable charger rental, 2) custom phone case with battery, 3) hallway charging station, 4) battery-buying buddy system, 5) cable lending program, 6) classroom outlet map, 7) low-power-mode tutorial, 8) backup juice pack subscription, 9) 'phone fasting' campaign, 10) charger-and-charge cafeteria service. Finalists: 1, 3, 8. Selected: 1 — needs lowest capital, fastest to test, and aligns with 'experience' lever (ease).*"

Scoring guide (3-2-1): - **3:** 10 distinct ideas, 3 finalists, 1 selected with criterion-based reasoning. - **2:** 8+ ideas, finalists named, weak criterion. - **1:** Fewer than 8 ideas, OR no criterion.

SA2. *"Read this short business case (provided): 'Bento Buddies cut its delivery cost from $1 to $0.40 per box by routing 6 boxes per stop.' Which value lever did Bento Buddies pull, and how?"*

Model answer: "*They pulled the LOWER PRICE lever — by reducing variable cost per unit from $6 to $5.40, they could either lower customer price or increase margin. They did it through 'better experience' for the delivery driver (route density), which compounded into the price lever.*"

Scoring guide (3-2-1): - **3:** Names the lever AND traces the operational change to the lever. - **2:** Names the lever, traces vaguely. - **1:** Names the wrong lever.

Performance Task: Ideation Pack (90 min over 2 sessions).

Child runs a SCAMPER + HMW ideation flow on a chosen problem, produces 10 ideas, narrows to 3 finalists with named criteria, picks 1, and writes a 1-paragraph defense. **Rubric:** Reflection Writing Rubric (Grade 8): (1) 10 distinct ideas, (2) 3 finalists screened against ≥3 criteria, (3) selection defended with criteria + customer link. **Strand outcome:** PSIG + BAVC.

Extension (advanced learner):

"*Of your 10 ideas, which would FAIL the fastest? Run a 5-minute premortem on the failure mode.*" Previews Grade 10 premortem.

Unit 2: The break-even truth

MC1. *"Break-even formula: units × price = fixed cost + variable cost × units. For a fixed cost of $40, price of $5, variable cost of $2 per unit — what is break-even units?"*

A. 8 units · B. 13 units · C. 14 units · D. 20 units

Answer: C · **Standard:** *CCSS-Math 8.EE.C.7 — Linear equations.* · **Strand outcome (MFL, 8):** *I can build a simple break-even calculator (units × price =*

fixed + variable × units) and use it to find break-even volume for a sample business. · **Misconception targeted:** "*Break-even is just sales = cost*" — corrects: 5u = 40 + 2u → 3u = 40 → u = 13.33, so 14 to fully cover.

MC2. *"Worked Example C: $0.20 price, $0.30 variable cost, $20 fixed cost. At what units do you break even?"*

A. 100 · B. 200 · C. There is no break-even — every cup loses money · D. 500

Answer: C · **Standard:** *CCSS-Math 8.F.B.4 — Construct functions.* · **Strand outcome (MFL, 8):** *I can build a simple break-even calculator and use it to find break-even volume for a sample business.* · **Misconception targeted:** "*If I sell more, I'll be profitable*" — corrects with negative-CM scenario.

MC3. *"Break-even is..."*

A. The goal · B. The floor — 'I stopped losing money' · C. The ceiling · D. The same as profit

Answer: B · **Standard:** *CCSS-Math 8.F.B.4.* · **Strand outcome (MFL, 8):** *I can build a simple break-even calculator.* · **Misconception targeted:** "*Break-even is the goal*" — corrects directly.

SA1. *"Build a break-even calculator for: $50 fixed cost, $4 price per unit, $1.50 variable cost. Show the formula AND the break-even units. Then show what happens at 30 units (profit or loss?)."*

Model answer: "*Formula: 4u = 50 + 1.5u → 2.5u = 50 → u = 20 units to break even. At 30 units: revenue = $120, cost = $50 + $45 = $95, profit = $25.*"

Scoring guide (3-2-1): - **3:** Break-even = 20 AND 30-unit profit = $25 with a clean computation. - **2:** Break-even right, 30-unit math wrong. - **1:** Break-even wrong.

SA2. *"Run a 1-variable sensitivity: keep break-even formula above, but raise variable cost from $1.50 to $2.00. New break-even units? What's the lesson for ops?"*

Model answer: "*4u = 50 + 2u → 2u = 50 → u = 25 units. A 33% rise in variable cost raises break-even by 25%. Lesson: every dollar of variable cost is multiplied across all units; even small material-price spikes meaningfully shift break-even.*"

Scoring guide (3-2-1): - **3:** New BE = 25 AND a generalized lesson. - **2:** New BE right, lesson vague. - **1:** Math wrong.

Performance Task: Break-Even Pack — Linear Calculator + Sensitivity Table (90 min).

Child builds a simple break-even calculator and a 2x2 sensitivity table varying price and variable cost; writes a 1-paragraph interpretation (which input the result is most sensitive to). **Rubric:** Budget Project Rubric (Grade 8): (1) calculator math correct, (2) sensitivity 2x2 populated, (3) interpretation names most-sensitive input with reasoning. **Strand outcome:** MFL + OSO.

Extension (advanced learner):

"*Your break-even is 20 units. Your venture sells 14 units a week. What 3 strategic moves could you make in the next 4 weeks (other than 'sell harder')?*" Targets ops-thinking; previews Grade 10 funding/structure.

Unit 3: Marketing as experiment

MC1. *"PDSA stands for..."*

A. Plan, Do, Study, Act · B. Pretty Darn Smart Always · C. Plan, Develop, Sell, Achieve · D. Practice, Drill, Score, Adjust

Answer: A · **Standard:** *NGSS-ETS MS-ETS1-3 — Analyze data from tests.* · **Strand outcome (LRCI, 8):** *I can complete a Plan–Do–Study–Act (PDSA) cycle on one part of my business and document the change, the result, and the next test.* · **Misconception targeted:** "*PDSA is just trying things*" — corrects with named steps.

MC2. *"In a 3-day A/B test, you collected 40 customer interactions split 22 vs. 18. Is this a clear winner?"*

A. Yes — 22 wins · B. No — direction-suggestive but small sample; could be noise; consider running more · C. Yes, exactly · D. Tied

Answer: B · **Standard:** *CCSS-Math 8.SP.A.1 — Patterns of association in bivariate data.* · **Strand outcome (CBM, 8):** *I can run a 3-day mini-marketing test (two posters, two messages) and report which performed better, by what measure, and what I'd do next.* · **Misconception targeted:** "*A/B test results give you a winner*" — corrects.

MC3. *"Variance between plan and actuals is..."*

A. A sign your plan was bad · B. Data — tells you something about reality and the plan · C. Something to hide · D. Random noise

Answer: B · **Standard:** *CCSS-ELA W.8.1 — Write arguments.* · **Strand outcome (OSO, 8):** *I can build a 4-week project plan with milestones, owner, and deadline; track actuals vs. plan; and write a 1-page variance explanation.* · **Misconception targeted:** "*Variance is a sign of failure*" — corrects.

SA1. *"Document a complete PDSA cycle for a marketing change you'd test. PLAN: hypothesis. DO: what you did. STUDY: what happened (with a number). ACT: next test."*

Model answer: "*PLAN: Hypothesis — adding a price tag ($1.50) to our flyer will increase customer-stops vs. the no-price version because hesitant buyers won't have to ask. DO: Ran flyer A (with price) and flyer B (without) for 3 days each, same hallway. STUDY: A drew 28 stops, B drew 21 stops — 33% lift in stops.*

Sample is small, so direction-suggestive. ACT: Next test — does adding price + 'limit 5 per customer' increase further? Run for 5 days for tighter signal."

Scoring guide (3-2-1): - **3:** All 4 steps named with specifics AND a sample-size hedge. - **2:** All 4 steps but 1 weak. - **1:** Steps missing.

SA2. *"Your 4-week project plan: Week 1 — set up booth (planned 3 hr, took 6 hr); Week 2 — first sales (planned 20 units, sold 12). Write a 1-paragraph variance explanation."*

Model answer: "*Two main variances: setup took 2x planned (6 vs. 3 hr) because the wholesale supplier mis-shipped and we waited 1.5 hr for replacement; this is a one-time delay. Sales hit 60% of plan (12 vs. 20 units); customer-interview data suggests our $4 price is high vs. the $2.50 willingness-to-pay among the soccer-team segment. The setup variance is bounded; the sales variance is structural and points to a price-test next week.*"

Scoring guide (3-2-1): - **3:** Both variances quantified, root cause named, action implied. - **2:** Variances named, reason for one is vague. - **1:** "Things went bad."

Performance Task: Marketing Test Pack + First PDSA Cycle (week-long).

Child runs a 3-day A/B marketing test with daily logs, completes a full PDSA cycle on the test, and writes a 1-page variance explanation against their plan. **Rubric:** Customer Research Rubric (Grade 8): (1) test design fair (controlled vars), (2) data tracked daily, (3) PDSA all four steps documented, (4) interpretation hedges sample size. **Strand outcome:** CBM + LRCI + OSO.

Extension (advanced learner):

"*At what sample size would you trust your A/B test result enough to fully commit? At 12 you didn't. At 100 maybe. What's your threshold rule?*" Targets statistical thinking; previews Grade 11 sales analytics.

Unit 4: Operate smarter

MC1. *"For a hypothetical kid- or teen-run online business, the BIGGEST privacy risk is..."*

A. The business name is too long · B. Collecting customer names + addresses without secure storage or a clear policy · C. The website is slow · D. The colors clash

Answer: B · **Standard:** *Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2 — Identify ways that consumers may avoid making poor decisions.* · **Strand outcome (ESL, 8):** *I can identify one privacy risk and one safety risk in a kid- or teen-run online business and write a 3-step mitigation plan for each.* · **Misconception targeted:** "*Privacy isn't real for small businesses*" — corrects.

MC2. *"Peer feedback round: 'What's working / What's confusing / One change I'd try' — the third prompt is best because..."*

A. It demands change · B. It's actionable — names a specific next step · C. It's polite · D. It's required

Answer: B · **Standard:** *CASEL Self-Management — Goal-setting, self-monitoring.* · **Strand outcome (LTO, 8):** *I can give and receive a structured peer feedback round using "What's working / What's confusing / One change I'd try."* · **Misconception targeted:** "*Vague feedback helps*" — corrects.

MC3. *"At PDSA Cycle 2, your variance is bigger than at Cycle 1. This means..."*

A. PDSA isn't working · B. Your hypothesis was wrong AND you have new info — that's progress · C. Quit · D. Rerun the same test

Answer: B · **Standard:** *NGSS-ETS MS-ETS1-3.* · **Strand outcome (LRCI, 8):** *I can complete a PDSA cycle on one part of my business and document the change, the result, and the next test.* · **Misconception targeted:** "*PDSA failure means project failure*" — corrects.

SA1. *"Write a 3-step mitigation plan for one privacy risk in a hypothetical kid-run online button-pin business."*

Model answer: "*Risk: customer names + shipping addresses stored in a shared Google Doc anyone with the link can edit. Step 1: Move data to a Doc with view/edit restricted to me + co-founder only; turn link-sharing OFF. Step 2: Delete records older than 90 days unless customer opts in; tell customers in writing. Step 3: For credit-card-like data, use a real platform (Stripe, Square) — never store card numbers in personal docs.*"

Scoring guide (3-2-1): - **3:** Specific risk + 3 specific actionable steps with order. - **2:** Risk + 3 steps, one vague. - **1:** Generic steps.

SA2. *"Run a peer feedback round on a teammate's pitch. Use 'What's working / What's confusing / One change I'd try.' Each line specific."*

Model answer: "*WORKING: Your '$3 a bracelet' price was clear and earned a confirming nod from 4 of 5 listeners — keep that line. CONFUSING: When you said 'eco-friendly,' I wasn't sure what was eco about it (recycled materials? compostable packaging?). Could you replace 'eco-friendly' with the specific thing? CHANGE: I'd try moving the customer story (Maya, the 7th-grader) to the start; right now it lands after the price.*"

Scoring guide (3-2-1): - **3:** All three prompts specific and constructive. - **2:** Two of three. - **1:** One specific.

Performance Task: Final PDSA Pack (4-week project).

Child runs a 4-week project plan with weekly milestones, owner, deadline, tracks variance, runs ONE complete PDSA cycle on a chosen part of the venture, and

writes a 1-page privacy/safety risk plan with mitigations. **Rubric:** Business Plan Rubric (Grade 8): (1) plan tracked weekly, (2) variance explained with root causes, (3) PDSA documented end-to-end, (4) risk plan specific with 3-step mitigations. **Strand outcome:** OSO + LRCI + ESL + LTO.

Extension (advanced learner):

"*Your team disagrees about the next PDSA cycle. Two members want to test pricing; one wants to test product quality. Write the criteria for choosing which test runs first.*" Anchors prioritization; previews Grade 11 backlog work.

Total this grade: 12 MC + 8 SA + 4 Performance Tasks + 4 Extensions = 28 items.